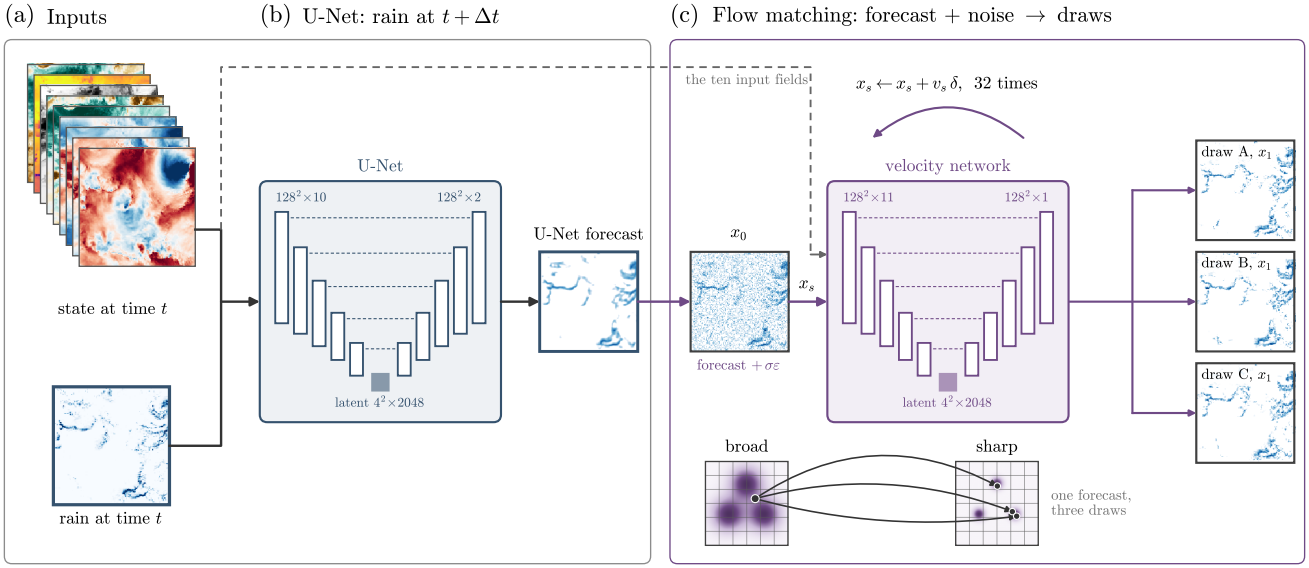

(a) Inputs
(b) U-Net: rain at $t + \Delta t$
(c) Flow matching: forecast + noise → draws
state at time t
rain at time t
U-Net
$128^2 \times 10$
$128^2 \times 2$
latent $4^2 \times 2048$
U-Net forecast
the ten input fields
$x_s \leftarrow x_s + v_s\,\delta$, 32 times
velocity network
$128^2 \times 11$
$128^2 \times 1$
latent $4^2 \times 2048$
x_0
x_s
forecast $+ \sigma\varepsilon$
draw A, x_1
draw B, x_1
draw C, x_1
broad
sharp
one forecast,
three draws